

Amaan Pathan

Backend Engineer | Golang · TypeScript · Node.js · PostgreSQL · Redis · Docker · Distributed Systems

+91-9910817022 • eulerbutcooler@gmail.com • Lucknow, India • github.com/eulerbutcooler • linkedin.com/in/amaan-pathan • eulerbutcooler.tech

SUMMARY

Backend Engineer with open source experience at BookBrainz and Guidefox and production freelance work at NIAT serving 500+ active users. Proficient in Golang, TypeScript/Node.js, PostgreSQL, Redis, and large-scale backend architecture. Delivered 3+ end-to-end backend systems including a multi-step workflow automation engine and a fully functional HTTP/1.1 server built from raw TCP sockets. Committed to clean architecture, API security, and measurable performance improvements.

SKILLS

- **Languages:** Golang, TypeScript, JavaScript (ES6+), Python, SQL
- **Backend:** RESTful APIs, Distributed Systems, System Design, JWT Authentication, RBAC, NATS JetStream, Workflow Automation, Microservices, Node.js, Express.js
- **Databases:** PostgreSQL, Redis, MongoDB; schema design, query optimization, connection pooling, Prisma ORM
- **Infrastructure & DevOps:** Docker, Linux/Unix, Bash, Nginx, Kubernetes, AWS, Cloudflare, CI/CD pipelines, Git, GitHub
- **Networking & Systems:** HTTP/1.1, TCP/IP, OS internals, concurrency, goroutines, connection lifecycle management
- **Tools & Practices:** Postman, VS Code, Zod, DTO-based API contracts, unit & integration testing

WORK EXPERIENCE

Open Source Contributor | BookBrainz | Remote

Jan. 2026 – Mar. 2026

- Replaced external Docker wait-scripts with Docker-native health checks across 3 services (PostgreSQL, Elasticsearch, Redis), eliminating startup race conditions and reducing environment setup failures by ~90%.
- Resolved GitHub Actions CI failures caused by Docker version incompatibilities on `ubuntu-latest` runners, restoring a 100% CI pass rate and unblocking the project's active PR review pipeline.

Software Engineer | NIAT (Freelance / Contract) | Remote

Sep. 2025 – Oct. 2025

- Designed RESTful backend services for an internal platform with 500+ active users; delivered authentication, RBAC, and a RAG pipeline improving query accuracy by 35%.
- Optimized API response times to under 120ms under concurrent load through query tuning and targeted caching strategies.

Open Source Contributor | Guidefox | Remote

Oct. 2024 – Jan. 2025

- Integrated Zod-based validation across 20+ backend APIs, reducing runtime errors by 40%; introduced DTO-based contracts cutting duplicated backend logic by 30%.
- Expanded test coverage from 55% to 80% through targeted unit and integration tests, cutting defect escape rate and accelerating the project's CI merge cycles.

PROJECTS

Hermes – Fault-Tolerant Workflow Automation Engine [GitHub](#) Golang · PostgreSQL · Redis · Docker · NATS JetStream

- Built a self-hostable event-driven workflow automation engine with parallel workers handling 1,000+ jobs/hour across multi-step automation pipelines.
- Enforced retries, idempotency, and failure isolation; used NATS JetStream to guarantee at-least-once delivery semantics.

HTTP Server from Scratch [GitHub](#) Golang · TCP Sockets · HTTP/1.1

- Engineered a fully functional HTTP/1.1 server from raw TCP sockets; handled concurrent connections with goroutines and complete connection lifecycle control.
- Benchmarked at 3,000 requests/second locally, validating low-level networking and OS fundamentals.

Wingman – AeroMentor Platform Backend [GitHub](#) Golang · PostgreSQL · NATS JetStream · Redis · MinIO · Qdrant · Docker

- Architected the Go backend for AeroMentor (an AI-powered LMS) following Clean Architecture with ports-and-adapters; shipped course, lesson, analytics, and JWT/RBAC endpoints with Redis token-bucket rate limiting at 60 req/min per user.
- Developed an async RAG document ingestion pipeline via NATS JetStream and Qdrant to power AI chat and quiz generation; handled file delivery through presigned MinIO URLs and TUS resumable uploads, eliminating file bytes from the server hot path.

EDUCATION

B.Tech in Computer Science Engineering | CGPA: 8.0 / 10.0

2022 – 2026

University of Lucknow, Lucknow, India

ACHIEVEMENTS

- Maintained 8.0 CGPA throughout B.Tech CSE while independently shipping 3+ production-grade backend systems and contributing to two active open source projects simultaneously.
- **LeetCode** – 270+ problems solved across Arrays, Trees, Dynamic Programming, Graphs, and Hashing; consistent practice across core data structures and algorithms.